



# AUTOMATION: The Holy Grail Of MODERN AGRICULTURE

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Centrally-controlled irrigation can improve productivity in fields

*This article discusses the various solutions for smart farming, and how India is responding to them*

Throughout the ages, yield from the farm has been majorly dependent on the manual effort put in by the agri community. External factors include weather, water supply and land condition. Introduction of automated equipment and sensorised devices for agriculture ensures that the yield keeps improving, even if other factors are not favourable. We look into the various technologies that are delivering great benefits, and how India is responding to these advancements.

## Automated irrigation

Provision for the correct amount of usable water to the farm is extremely crucial. It not only keeps production consistent, it also paves the way to generate a variety of crops. Irrigation methods are many, such as surface irrigation, micro-irrigation, drip irrigation, sprinkler irrigation and more. In all cases, the focus should be on the amount of water being distributed.

Satish K.S., founder, Flybird Farm Innovations Pvt Ltd, says, "Small- and marginal-size farmers typically cannot afford irrigation controllers and have to manually control water flow, which can be inaccurate and inconsistent. Over-irrigation or under-irrigation can greatly damage the crops."

Excess water causes damage due to water-logging in soil, while insufficient water can dry out the crops or result in underwhelming yield. There can be collateral damage to such conditions as well, like sub-optimal water utilisation and water wastage, manifestation of insects and pests, damaged soil conditions—all of which add up to major financial losses for farm owners.

This is where automation plays a vital role. Automated irrigation keeps the soil's moisture level optimum with no deviations. How is this achieved? Parth Bhatt, product specialist, Energy Bots Pvt Ltd, explains, "An automated setup first understands the wetness of the soil. Strategically-placed humidity, moisture and weather sensors (for rainfall, snowfall and other such conditions) analyse the soil moisture content. Based on a certain threshold value, if the moisture level is measured lower, the microcontroller signals the water line to automatically start the water supply.

"On the other hand, if moisture content exceeds the threshold, water supply is instantly stopped."

This mechanism ensures that the fields are optimally moisturised—neither too much, nor too little. It also saves misuse of

